

Desugaring

CS 350

Dr. Joseph Eremondi

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 - Otherwise we'd just code in machine code/assembly all the time
- These “nice to have” are called *syntactic sugar*
 - They “sweeten” the experience of programming

Example of Sugar

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    f();  
}  
// same as  
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 - Simplifies the core language

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- Add translation from surface to core AST

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 - The type of tree we give to `interp` to actually run

Subtraction: Datatype

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```
(define-type ExpS
  (numS [n : Number])
  (plusS [left : ExpS]
         [right : ExpS])
  (timesS [left : ExpS]
          [right : ExpS])
  (cndS [test : ExpS]
        [thenBranch : ExpS]
        [elseBranch : ExpS])
  (subS [left : ExpS]
        [right : ExpS]))
```

- Exact same as core AST, but with subtraction

Subtraction: Parser Case

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```
(define (parse [s : S-Exp]) : ExpS
  ...
  [(s-exp-match? `{- ANY ANY} s)
   (subS (parse (second (s-exp->list s)))
          (parse (third (s-exp->list s))))])
```

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(define (desugar [es : ExpS]) : Exp
  (type-case ExpS es
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     (numE n)]
    [(boolS b)
     (boolE b)]
    [(plusS l r)
     (plusE (desugar l) (desugar r))]
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 - Recursively desugar the parts
 - Re-assemble with the corresponding Exp constructor

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- Example:

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(define (subE [l : Exp]  
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  : Exp  
  (plusE l  
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- Note: subE is a function, *not* a constructor

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 - e.g. what if `-` occurs deep in an expression

More Examples: Equality

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```
(define-type ExpS
  ...
  (eqS [l : ExpS] [r : ExpS]))

(define (eqE [l : Exp]
             [r : Exp])
  : Exp
  (zero?E (subE l r)))

(define (desugar [es : ExpS]) : Exp
  ...
  [(eqS l r)
   (eqE (desugar l) (desugar r))])
```

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(define (andE [l : Exp]  
              [r : Exp])
```

```
  : Exp
```

```
  (cndE l  
        r  
        (boolE #f)))
```

```
(define (orE [l : Exp]  
             [r : Exp])  
  (cndE l (boolE #t) r))
```

```
(define (notE [e : Exp])  
  (cndE e (boolE #f) boolE #t))
```

- See Curly-Desugar in URCourses for full implementation

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 - `interp` is usually the most complicated part

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 - Translated a small language into an even smaller one
- Compilation is just a bunch of desugaring passes
 - Simpler and simpler languages until we have something simple enough for assembly code

Example

- $\{\text{if } \{= \{+ 3 3\} 3\} 4 5\}$

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